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ODDEND SEMESTER EXAMINATION December- 2025

Name of the Course:

B.Tech Semester: V

Name of the paper: Design of Machine Elements- I

Paper Code: TME 502

Time: 3 hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1. 20 marks	
(a) Define mechanical properties and explain their significance in material selection.	CO1, CO5
(b) Write short notes on cast iron, alloy steels, and their mechanical applications.	
(c) What do you understand by "Phases of Design"? Explain with examples from machine components.	
Q2. 20 marks	
(a) A cantilever beam of circular cross-section is subjected to a concentrated load at its free end. Derive the bending stress formula.	CO2, CO4
(b) What is stress concentration? How do fillets and notches affect the design of machine components?	
(c) Explain the concepts of failure theories: Maximum Principal Stress and Distortion Energy theories.	
Q3. 20 marks	
(a) (a) What is cumulative fatigue damage? Explain in details	CO3,
(b) Derive the corrected endurance limit considering surface, size, and reliability factors.	CO2
(c) A component is subjected to fluctuating tensile stress varying between 50 MPa to 150 MPa. Plot the stresses on a Goodman diagram and state whether it is safe.	
Q4. 20 marks	
(a) Design a hollow shaft of 1.5 m length to transmit 200 kW at 250 rpm. Maximum shear stress = 60 MPa. Internal diameter to be 0.5 × external diameter.	CO4, CO1
(b) Discuss BIS specifications relevant to shaft design and failure considerations.	
(c) Explain the failure of a key in a key-shaft assembly and derive the relation for torque transmission.	
Q5. 20 marks	
(a) (a) Design a double-riveted butt joint with two cover plates for boiler plates of 12 mm thickness.	CO5,

(b)	Explain the analysis of eccentrically loaded welded joints subjected to bending moment.	CO6
(c)	Explain the design steps for a bolt in a flange coupling under initial tension and shear load.	